Project Report

On

Load Profile Management and Forecasting

An AI-Integrated Load Intelligence Dashboard for the Delhi Distribution Grid

The Project report is submitted for the partial fulfilment of the requirement for the degree of Bachelor of Technology in Electrical Engineering from the Faculty of Technology, University of Delhi.

Submitted by

Vansh Sachan [24295918040]

Animish Gupta [23295919010]

Ayush Rathore [24295918024]

Lavish Rawat [23295918041]

Under the guidance of

Dr. Ujjal Sur (Supervisor)

Dr. Arjun Tyagi (Co-Supervisor)

DEPARTMENT OF ELECTRICAL ENGINEERING

FACULTY OF TECHNOLOGY

UNIVERSITY OF DELHI

NORTH CAMPUS, DELHI-110007

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our supervisor, Dr. Ujjal Sur, and co-supervisor, Dr. Arjun Tyagi, for their invaluable guidance, encouragement and continuous support throughout the course of this project. Their insights into power system engineering, and their feedback at every stage, have been instrumental in shaping this work.

We are also thankful to the Department of Electrical Engineering, Faculty of Technology, University of Delhi, for providing us with the resources, laboratory facilities and academic environment necessary to carry out this project.

Finally, we would like to thank our families and friends for their constant encouragement and support.

Vansh Sachan · Animish Gupta · Ayush Rathore · Lavish Rawat

CONTENTS

Abstract 4
1. Introduction and Problem Statement 4
2. Objectives 4
3. Methodology 4
4. Mathematical Model 5
5. Results and Model Validation 5
6. Dashboard: 6 Live Pages, 6 Future Scope 7
7. Dataset 8
8. Novelty and Limitations 9
9. Work Distribution 9
10. Conclusion 9
References 9

Abstract

Electricity demand in Delhi is rising and becoming harder to predict because of humid heatwaves, rooftop solar and changing consumer behaviour. This project builds a load profile management and forecasting dashboard that combines power-system engineering with machine learning. Real Delhi SLDC load (1 April 2023 – 12 January 2026, 15-minute blocks) and Open-Meteo weather are used to characterise the load profile and to forecast all 96 blocks of the next day with a P10–P90 band. Six dashboard pages are live (Overview, Load Analytics, Weather Impact, Forecast, DSM ₹ Calculator and Transformer Health) and six are future scope. On the test window (13 July 2025 – 12 January 2026) the final multi-horizon LightGBM model reaches a test accuracy of ≈ 96% (MAPE 3.93%) against a MAPE of 5.16% for a seasonal-naive baseline, and its 80% conformal band covers 84.4% of test blocks. Load rises by about 112.0 MW per °C of heat index (r^2 = 0.61). The DSM and transformer pages are illustrative calculators, not regulatory figures. The system is built entirely on free and open-source tools.

1. Introduction and Problem Statement

India met an all-time peak demand of 256.1 GW on 25 April 2026 [1], and Delhi recorded its highest-ever peak of 8,748 MW on 29 June 2026, surpassing the previous record of 8,656 MW set in 2024 [2]. A 2025 study by the Centre for Science and Environment found that the heat index, which combines temperature and humidity, explains about 67% of the variation in Delhi's daily peak demand [3]. That figure is CSE's measure of daily-peak variation; our own regression of load on heat index over all 15-minute blocks (Section 4) gives r^2 = 0.61, a different metric on different data that is not directly comparable. At the same time, growing solar generation has reshaped the daily net-load curve into a "duck" shape, with midday exchange prices near ₹1.11 per unit and night prices near ₹9.71 per unit in May 2026 [4].

Problem statement: Distribution companies (DISCOMs) must schedule tomorrow's power in 96 blocks of 15 minutes. Any deviation from this schedule is charged under the CERC Deviation Settlement Mechanism Regulations, 2024 [5]. Existing monitoring tools show only past consumption; they do not tell an engineer what tomorrow's forecast error will cost, which transformer will overload in the heat, or how steep the evening ramp will be. This project addresses that gap.

2. Objectives

- Characterise Delhi's load profile with load factor, peak-to-average ratio and the load-duration curve.
- Quantify the effect of heat index (temperature plus humidity) on Delhi's load.
- Forecast the next day's 96 blocks with a calibrated P10–P90 band, and keep a model only if it beats a simpler baseline.
- Convert a deviation into rupees (DSM, illustrative) and a hot-spot temperature into transformer ageing (IEC 60076-7).
- Present every result in a dashboard where each number is labelled REAL, SYNTHETIC or ILLUSTRATIVE.

3. Methodology

3.1 System workflow

The system has five stages. (1) Delhi SLDC 5-minute load and Open-Meteo hourly weather are collected. (2) The load is cleaned causally, converted to IST, resampled to 15-minute blocks and joined with weather, a heat index and holiday and IPL calendars (Section 7). (3) A forecast engine predicts the next day's 96 blocks with a P10–P90 band. (4) Formula engines turn the data and forecast into a peak alert, load factor, heat sensitivity, an illustrative DSM charge and transformer ageing (Section 4). (5) A FastAPI service serves these results to a Next.js dashboard. IEX prices, ToD bills, solar net load, losses and theft detection belong to the future-scope pages (Section 6).

3.2 Models and validation protocol

Models are compared in increasing order of complexity on the same test period, and a more complex model is kept only if it beats the simpler one. The baseline is seasonal-naive (the same 15-minute block on the previous day). The main model is a single multi-horizon LightGBM that forecasts all 96 blocks with the horizon as a feature (37 features), tuned with Optuna (25 trials on validation MAPE) and early-stopped on validation. A SARIMAX benchmark on hourly data is implemented, but only its order search on validation was run (validation MAPE about 25%); it has no test score and is not in the results. N-HiTS, TFT and Chronos [12] were proposal-stage ideas and were not run. Split-conformal prediction (MAPIE), calibrated on validation, gives the P10–P90 band, an 80% central interval [11], [14]; TreeSHAP explains individual days. The data are split chronologically and never shuffled: train 1 Apr 2023 – 12 Jan 2025, validation 13 Jan – 12 Jul 2025, test 13 Jul 2025 – 12 Jan 2026 (183 days). Every forecast origin and horizon in the test window is scored (1,672,080 origin × horizon pairs), with MAPE [8] as the headline metric. A leakage audit checked every lag feature against the forecast origin.

4. Mathematical Model

Table 1 lists the equations behind the six live pages, with the same names and worked examples as the presentation. The formulas are the ones implemented in the project's engine code (ml/src/lpmf/engines); their 34 unit tests, including the worked examples below, pass.

Table 1: Equations of the six live pages, with this project's numbers

Page · equation	Formula	What it computes · worked example
Overview **Peak & overload alert**	$\hat{P}_{peak} = \max_t \hat{P}(t), \quad alert\ if\ \hat{P} > P_{limit}$	Highest forecast load of the day and when it occurs. 12 Jan 2026: block 42 → (42 − 1) × 15 min = 10:15 IST; 5,670.83 MW vs limit 8,748 MW → Clear (3,077 MW headroom). REAL.
Load Analytics **Load factor & PAR**	$LF = \frac{P_{avg}}{P_{max}}, \quad PAR = \frac{1}{LF}$	Average ÷ peak load; PAR = 1/LF [6]. Illustrative day (530 / 820): LF = 0.646, PAR = 1.55. Whole period, measured (Apr 2023 – Jan 2026, 96,398 blocks): LF = 4,316 / 8,582 = 0.503, PAR = 1.99.
Weather Impact **Heat sensitivity (all 15-min blocks)**	$\Delta P \approx \beta \cdot \Delta HI, \quad HI = f(T, RH)$	Extra MW per °C of heat index, OLS on 96,398 blocks [15]: β = 112.0 MW/°C, r^2 = 0.61; +3 °C → +336 MW. CSE's 67% [3] is for daily peaks, a different metric. REAL.
Forecast **Forecast accuracy (MAPE)**	$MAPE = \frac{100}{n} \sum \left\lvert \frac{y_i - \hat{y}_i}{y_i} \right\rvert$	Average error as a % of actual load [8]. Test MAPE 3.93% vs seasonal-naive 5.16% (full test set). Derived arithmetic, not measured: 4,500 MW × 3.93% ≈ 177 MW. REAL.
Forecast **Band coverage**	$Coverage = \frac{N(P_{10} \le y \le P_{90})}{n}$	Share of actuals inside the P10–P90 band [14]: 84.4% on test vs the 80% target. REAL.
DSM ₹ Calculator **DSM deviation charge**	$C_b = \min(\lvert D_b \rvert, V_L)\frac{NR}{4} + \max(\lvert D_b \rvert - V_L, 0)\frac{kNR}{4}$	Deviation within the limit at rate NR, excess at k × NR [5]. D = 60 MW, VL = 41 MW, NR = ₹5,000/MWh, k = 1.5: ₹86,875 per block, ₹3,47,500 = ₹3.475 lakh/hr. ILLUSTRATIVE.
Transformer Health **Transformer ageing**	$V = 2^{(\theta_h - 98)/6}, \quad L = \sum V_n \Delta t$	Ageing rate V doubles every 6 °C above 98 °C [10]. 40 °C ambient, full load: θh = 40 + 55 + 23 = 118 °C, V = 2^(20/6) = 10.08×; 1 h uses 10.08 h of life. ILLUSTRATIVE.

5. Results and Model Validation

5.1 Forecast accuracy

On the full test set the final model's MAPE is 3.93% against 5.16% for seasonal-naive, a 24% reduction in error (naive scored on the 1,656,240 test rows that have a forecast from the previous day). Error rises with lead time: 3.21% at h = 1–4, 3.48% at h = 5–24, 3.84% at h = 25–48 and 4.22% at h = 49–96, and the model beats seasonal-naive at every horizon (Figure 2). The 80% band covers 84.4% of test blocks with a constant width of 602 MW. Table 2 shows how the model reached this figure; it is scored on the 1,267,749 test rows that are real readings in every data version, which is why naive reads 5.22% there and 5.16% on the full test set.

Table 2: Model ladder on the common real test rows (n = 1,267,749)

Model	What changed	Test MAPE
Seasonal-naive	Baseline: the same 15-minute block on the previous day	5.22%
First LightGBM	Load features aligned to the origin; lost to naive at h = 74–96	4.93%
Causal imputer + target-aligned lags	Imputer no longer uses future data; lags aligned to the target	4.00%
Final model	Causal Hampel outlier filter; profile fill from past days only	3.92%
Final model, full test set	All 1,672,080 origin × horizon pairs (naive on the full set: 5.16%)	3.93%

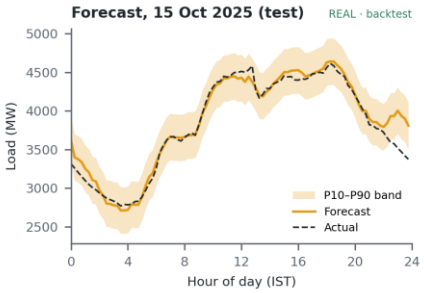


Figure 1: *Day-ahead forecast for 15 Oct 2025 (test), issued 23:45 the evening before; day MAPE 2.24%. REAL.*

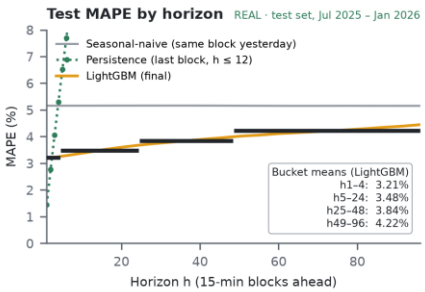


Figure 2: *Test MAPE by horizon vs seasonal-naive and persistence; black bars are the bucket means. REAL.*

Ablation. A 2×2(+1) factorial ablation (data version × feature set, fixed hyperparameters, 1,279,461 common test rows) separates the gain from 4.96% to 4.02% (−0.94 percentage points, 95% CI [−1.20, −0.68]). Most of it came from the target-aligned lags (Table 3). Removing the leak improved the score because the old imputer had inflated October–November late-night training labels by about 25%.

Table 3: Attribution of the −0.94 pp step (Shapley averages; re-tuning is the remainder)

Source of improvement	Share of −0.94 pp	pp
Target-aligned lag features	~64% (54–74% depending on order)	~0.61
Causal cleaning (imputer v1)	~25%	~0.23
Hyperparameter re-tuning	~10%	~0.10

5.2 Accuracy and Fit Check (Overfitting / Underfitting)

Table 4: Final-model accuracy by split (accuracy = 100 − MAPE; gain = 1 − model MAPE ÷ naive MAPE)

Split (rows)	MAPE	Accuracy	Seasonal-naive MAPE	Gain over naive
Train (5.91 M)	2.29%	97.71%	5.44%	58%
Validation (1.66 M)	4.29%	95.71%	5.97%	28%
Test (1.67 M)	**3.93%**	**96.07%**	**5.16%**	**24%**

Overfitting: not harmful. The train–test gap is real (2.29% vs 3.93%; Table 4, Figure 3) but small: early stopping kept 436 of 3,000 rounds, and validation and test error stay flat while only train error keeps falling. tlag_1day, the strongest feature, is not a leak: permuting it raises test MAPE from 3.93% to 23.4%.

Underfitting: no. The model beats seasonal-naive by 24% on test, at every horizon (3.21%, 3.48%, 3.84% and 4.22% for h = 1–4, 5–24, 25–48 and 49–96, against about 5.16%).

Why validation is higher. Two-thirds of validation is March–June heat, and seasonal-naive also errs more there (5.97% vs 5.16%).

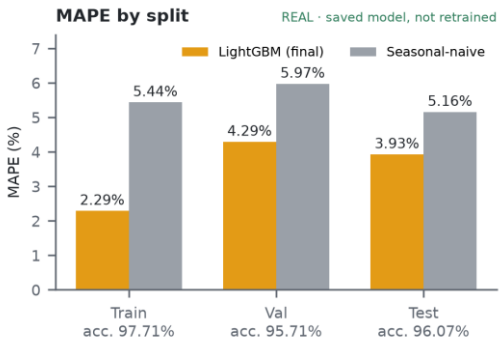


Figure 3: *MAPE by split, LightGBM vs seasonal-naive (saved model, not retrained); accuracy = 100 − MAPE. REAL.*

5.3 Load characteristics, heat sensitivity and ageing

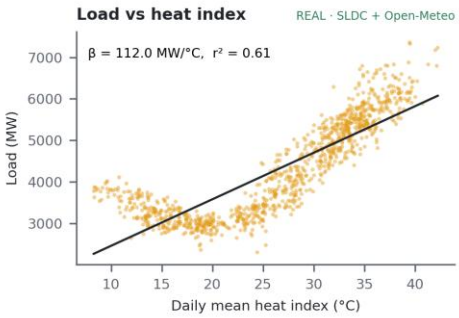


Figure 4: *Load vs heat index: 1015 daily means; line = OLS fit on all 15-min blocks. REAL.*

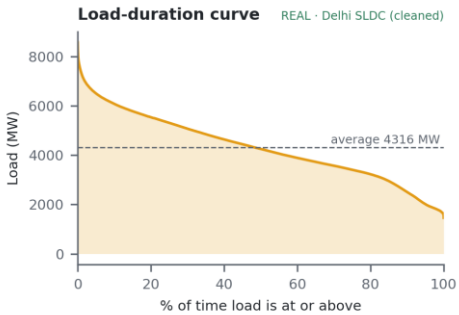


Figure 5: *Load-duration curve, 96,398 blocks, Apr 2023 – Jan 2026. REAL.*

Over the whole period the load factor is 0.503 and PAR 1.99 (average 4,316 MW, peak 8,582 MW on 19 Jun 2024). The duration curve (Figure 5) has a short steep tip: the highest loads last only a small share of the time. An OLS fit over 96,398 blocks gives β = 112.0 MW/°C and r^2 = 0.61 (Figure 4). Figure 6 shows the IEC 60076-7 ageing rate, a straight line on a log scale because ageing doubles every 6 °C.

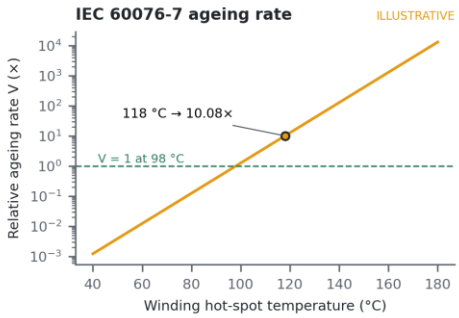


Figure 6: *IEC 60076-7 relative ageing rate; 118 °C → 10.08×. ILLUSTRATIVE.*

6. Dashboard: 6 Live Pages, 6 Future Scope

The dashboard has twelve pages, each answering one operational question (Table 5). Six are live and run on the API; six are future scope. Every chart and number carries a provenance badge (REAL, SYNTHETIC or ILLUSTRATIVE). The stack is Python (pandas, LightGBM, Optuna, MAPIE, SHAP), FastAPI with Pydantic response schemas, and Next.js 15 with React and Recharts.

Table 5: The twelve dashboard pages

#	Page	What it shows	Status
01	Overview	Day-ahead peak, peak time, alert (latest test day)	LIVE
02	Load Analytics	Duration curve, LF, PAR; load curve, heatmap planned	LIVE
03	Weather Impact	Load vs heat index; what-if slider planned	LIVE
04	Forecast	96 blocks, P10–P90 band, model comparison, SHAP (selected days)	LIVE
05	DSM ₹ Calculator	Illustrative charge for a given deviation	LIVE
06	Transformer Health	What-if: hot-spot temperature, ageing, life used per hour held	LIVE
07	IEX Price	Expensive blocks, purchase cost profile	FUTURE SCOPE
08	ToD Bill Simulator	Flat vs ToD vs optimised bill, shift advice	FUTURE SCOPE
09	Solar & Net Load	Duck curve, evening ramp, voltage risk	FUTURE SCOPE
10	Losses & PF	I^2R loss, kVAR needed, phase imbalance	FUTURE SCOPE
11	Theft & Segments	Suspicious meters, consumer load-shape groups	FUTURE SCOPE
12	Data & Model Card	Source, real vs synthetic, accuracy, limits	FUTURE SCOPE



Figure 7: *Overview page (live).*

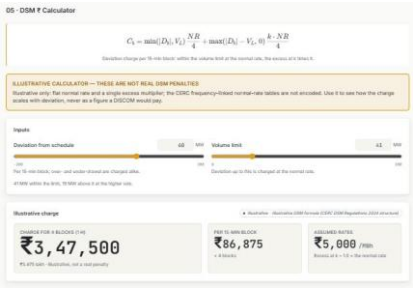


Figure 8: *DSM ₹ Calculator page (live).*



Figure 9: *Forecast page (live).*

7. Dataset

Every number on the six live pages comes from the files below. REAL = measured (Delhi SLDC, Open-Meteo); SYNTHETIC = generated stand-in; ILLUSTRATIVE = a worked example or user what-if. Column names are the exact fields in data/processed/delhi_15min.parquet, the feature file, the model outputs and the API schemas.

Table 6: Variables, sources and provenance

Variable(s)	Meaning	Source	Tag
`ts_ist, date, block`	15-min timestamp (IST), day, block 1–96	Delhi SLDC (Kaggle mirror)	**REAL**
`load_mw`	Delhi load per block (mean of three 5-min readings)	Delhi SLDC, cleaned	**REAL**
`imputed, impute_method, is_outlier`	Cleaning flags for every filled or rejected block	Cleaning pipeline	**REAL**
`temp_c, rh_pct, dewpoint_c`	Air temperature, relative humidity, dew point	Open-Meteo archive (28.6139 N, 77.2090 E)	**REAL**
`ghi_wm2, wind_ms, cloud_pct`	Irradiance, wind speed, cloud cover	Open-Meteo archive	**REAL**
`heat_index_c`	Heat index from temp_c and rh_pct (Rothfusz)	Derived from Open-Meteo	**REAL**
`is_holiday, holiday_name, dow, month, is_weekend`	Calendar features	Holiday calendar	**REAL**
`is_ipl_season`	IPL match season flag	ipl_seasons.csv	**REAL**
`tlag_1day, tlag_4w_avg, load_roll_mean_24h, night_temp_c`	Main model features (lags, rolling mean, night temperature)	Built from the above	**REAL**
`predicted_mw, p10_mw, p90_mw, actual_mw, ts_target`	Forecast, band and actual per test block	LightGBM + MAPIE backtest	**REAL**
`peak_mw, peak_time, peak_block, limit_mw, alert`	Overview outputs (limit = 8,748 MW record)	Engine on forecast	**REAL**
`load_factor, par, p_avg_mw, p_max_mw`	Load Analytics outputs	Engine on load_mw	**REAL**
`beta_mw_per_c, intercept_mw, r_squared, n_blocks`	Weather Impact outputs	Engine on load_mw, heat_index_c	**REAL**
`deviation_mw, schedule_limit_mw, normal_rate_per_mwh, excess_multiplier`	DSM ₹ Calculator inputs → penalty_rupees	User what-if input	**ILLUSTRATIVE**
`hotspot_temp_c, hours`	Transformer Health inputs → ageing_rate_v, life_hours_used	User what-if input	**ILLUSTRATIVE**
`category, load_mw (5 files)`	Residential, commercial, industrial, agricultural, EV profiles	Parametric generator	**SYNTHETIC**

7.1 Cleaning steps (all causal: each decision uses only readings at or before it)

- Drop duplicate timestamps; reindex onto a complete 5-minute grid (293,184 slots expected, 285,252 present: 2.705% missing).
- Reject physically impossible values: load ≤ 0, above 12,000 MW, or a jump above 800 MW in 5 minutes.
- Trailing Hampel filter (35-min window, 3σ) flags outliers without looking at future readings: 249 blocks flagged.
- Tiered gap-filling by gap length: short gaps forward-filled (894 blocks), medium gaps from the time-of-day profile of the past 5 days (1,153 blocks); gaps over 3 hours left missing, never fabricated.
- Resample to 15-minute blocks (block 1–96), join weather and calendars, and assert no duplicate or missing timestamps.

Table 7: Row counts (15-minute blocks)

Set	Period	Blocks	Usable (non-missing)
All data	1 Apr 2023 – 12 Jan 2026	97,728	96,398 (1,330 = 1.361% left missing)
Train	1 Apr 2023 – 12 Jan 2025	62,687	61,620
Validation	13 Jan 2025 – 12 Jul 2025	17,376	17,312
Test	13 Jul 2025 – 12 Jan 2026	17,665	17,466
Synthetic profiles	Same period, 5 categories	97,728 each	Future-scope pages only

Synthetic data. The five category profiles are built by a parametric model: base + daily pattern + weekly pattern + heat effect + noise + event bumps, validated against the real aggregate shape (peak timing, load-duration curve, autocorrelation). They serve only the future-scope pages; the six live pages run on real data, except the two illustrative calculators.

8. Novelty and Limitations

Novelty: Delhi load forecasting, theft detection and ToD calculators each exist separately in open-source work. We did not find an open project that links one Delhi forecast to a DSM charge under CERC 2024 and DERC limits, transformer ageing under IEC 60076-7 and, by design, ToD billing, solar ramps and theft ranking in one dashboard with every number labelled by provenance. The integration on Indian regulation is the contribution; the first six pages are built, the rest is future scope.

Limitations: we report only our own backtest, and it has these limits.

- **Test set not fully untouched:** the target-aligned lags were designed after reading the first model's test-set errors. Tuning and early stopping never saw test, but 3.93% may be slightly optimistic.
- **Shortest horizons:** persistence (next block = last block) beats the model at h = 1 (1.44% vs 3.18%) and h = 2 (2.77% vs 3.20%). The model beats seasonal-naive at every horizon.
- **Holidays:** on Diwali (20 Oct 2025) day MAPE was 8.65% and only 49% of blocks fell in the band; actual load was about 2,000 MW below the 4-week same-block average. Holidays are learned from few examples through one indicator.
- **Date range:** the data end on 12 Jan 2026, so Delhi's record 8,748 MW (29 Jun 2026) is not in the data (dataset maximum 8,581.6 MW, 19 Jun 2024), and the test window has no March–June summer.
- **Marginal coverage:** 84.4% is an average; by month it ranges from 73% (August) to 98% (November), and by horizon from 90% (h = 1–4) to 82% (h = 49–96).
- **Not run or illustrative:** SARIMAX has no test score; N-HiTS, TFT and Chronos were not run. DSM rupees use a flat rate (CERC frequency-linked tables not encoded); the transformer page is a what-if.

9. Work Distribution

Table 8: Work distribution among team members

Member	Roll No.	Contribution
Vansh Sachan (Group Leader)	24295918040	Model training and validation (naive, SARIMAX, LightGBM, leakage audit, ablation); built the live dashboard (FastAPI, formula engines, Next.js, 6 pages); data acquisition (SLDC, Open-Meteo); system architecture.
Animish Gupta	23295919010	Data refinement and cleaning (causal outlier filtering, gap-filling, final train/val/test feature set); author of this report.
Ayush Rathore	24295918024	Data refinement and cleaning (feature engineering, calendar/weather joins, final training data); prepared the presentation.
Lavish Rawat	23295918041	Data refinement and cleaning (data validation, QA checks, dataset documentation); prepared the presentation, references and formatting.

10. Conclusion

The project goes beyond a conventional load forecast by carrying a real Delhi forecast to its consequences: a peak alert, the shape of demand, its heat sensitivity, an illustrative DSM charge and transformer ageing. Six pages are live on real data, with every number labelled. The forecasting core reaches a test accuracy of ≈ 96% (MAPE 3.93%; seasonal-naive 5.16%) with an 80% band that covers 84.4% of test blocks. Holiday behaviour, the shortest horizons and the untested pre-monsoon summer peak are the main gaps, and the six future-scope pages are the next step.

References

[1] Ministry of Power (PIB), "India meets all-time highest peak demand of 256.1 GW," Apr. 2026.

[2] Delhi SLDC, peak demand record of 8,748 MW, 29 Jun. 2026 (reported by Business Standard).

[3] Centre for Science and Environment, "How Delhi is losing its cool," 2025.

[4] S. Sanyal and S. Dev, "The Duck and the Camel: Tracing the Net Load on the Indian Power Grid," EAC-PM Working Paper 53/2026.

[5] CERC (Deviation Settlement Mechanism and Related Matters) Regulations, 2024; DERC intra-state DSM order, 13 Sep. 2024.

[6] V.K. Mehta and R. Mehta, Principles of Power System, S. Chand (Ch. 3, Ch. 6).

[7] T. Gönen, Electric Power Distribution Engineering, 3rd ed., CRC Press, 2014 (Ch. 2).

[8] R.J. Hyndman and G. Athanasopoulos, Forecasting: Principles and Practice, 3rd ed., OTexts, 2021.

[9] Ministry of Power, Electricity (Rights of Consumers) Amendment Rules, 2023.

[10] IEC 60076-7:2018, Loading guide for mineral-oil-immersed power transformers.

[11] T. Hong and S. Fan, "Probabilistic electric load forecasting: A tutorial review," Int. J. Forecasting, vol. 32, no. 3, pp. 914–938, 2016.

[12] A.F. Ansari et al., "Chronos: Learning the Language of Time Series," arXiv:2403.07815, 2024.

[13] D.P. Kothari and I.J. Nagrath, Modern Power System Analysis, McGraw-Hill (Ch. 5).

[14] A.N. Angelopoulos and S. Bates, "A Gentle Introduction to Conformal Prediction," arXiv:2107.07511, 2021.

[15] L.P. Rothfusz, "The Heat Index Equation," NWS Technical Attachment SR 90-23, 1990.